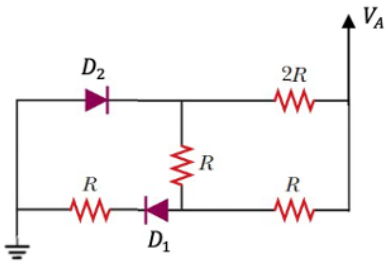
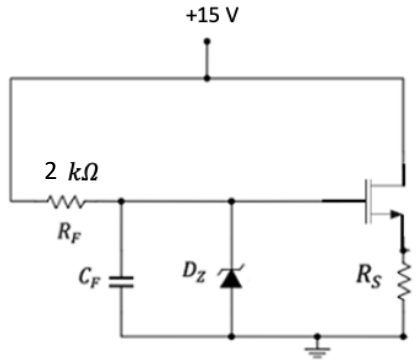
For the circuit below, calculate the power delivered to each diode and record in the table. For both diodes $V_{DO} = 0.7$ V. Show all steps.

V_A	R	P_{D1}	P_{D2}
+10 V	2 $k\Omega$		
-10 V	2 $k\Omega$		



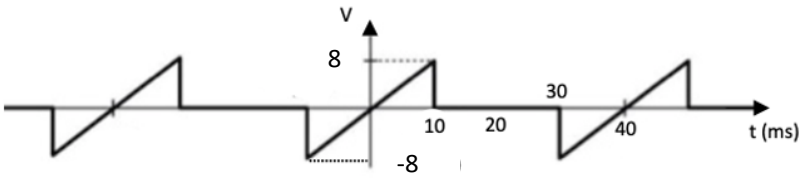
The circuit below has been connected for a long time. The Zener diode has $V_{zo} = 5.3$ V, $r_z = 10\ \Omega$ and $I_{z_min} = 0.5$ mA. The NMOS has $V_t = 1$ V, k=1 mA/V^2, and λ=0. Take $R_S = 1\ k\Omega$ and $C_F = 100\ \mu F$.

a) Calculate the current in resistor R_F.
b) Calculate the current in resistor R_S.
c) What is the effect of doubling the value of R_F on the operation of the circuit? Show.

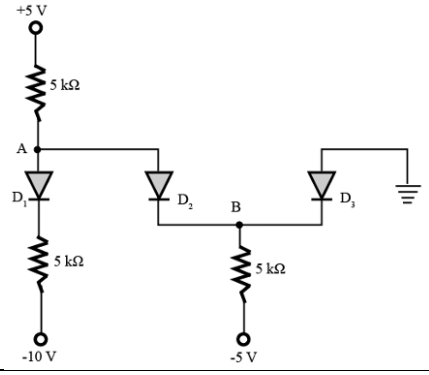


Consider the standard bridge rectifier with a load resistor of 2 $k\Omega$ connected to it. The input to the rectifier is the signal shown below. For all diodes $V_{DO} = 0.8$ V.

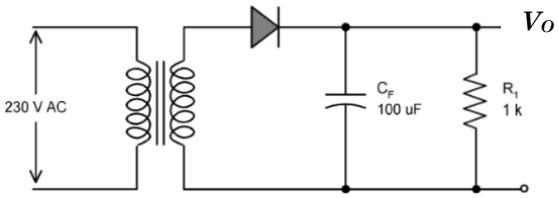
a) Draw the rectified output voltage waveform. Label all important points.
b) Find the average rectified output voltage and the average load current.
c) If a capacitor of 470 μF is connected to the rectifier, draw the output waveform for this case (on top of the figure in part a). Show all details.



Find the value of the current through diodes D_1 , D_2 and D_3. Use ideal diode model.

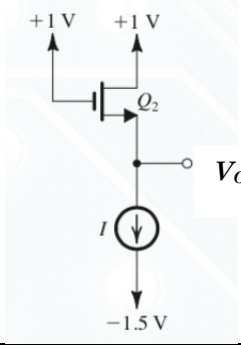


The diode in the circuit below has a forward drop voltage of 0.8 V. The ac line voltage is sinusoidal with an amplitude of 230 V at 60 Hz. The transformer has a 15:1 ratio.



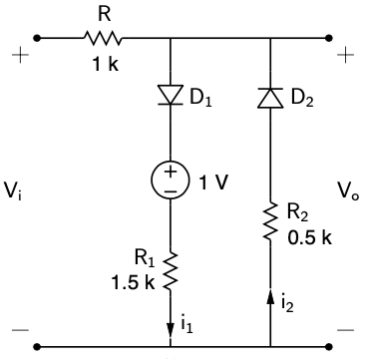
a) Draw the output voltage waveform. Label all important points in the vertical and horizontal axes.
b) Find the average output voltage and the average load current.
c) If a Zener diode with V_{zo} = 8.5 V, r_z = 0 Ω is connected parallel to the capacitor, what is the new value of the average output voltage. What will happen to this circuit?

In the circuit below, the transistor has $|V_t|$=0.5 V, k=1 mA/V^2, and λ=0. If I =0.2 mA, find the voltage V_o.



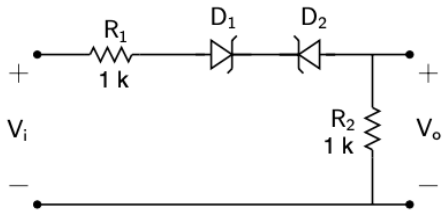
For the circuit below, find the indicated currents and voltage and record in the table. For all diodes V_{DO} = 0.7 V. Show all steps.

V_i	I_1	I_2	V_o
+1.5 V			
+ 2.5 V			
-5 V			



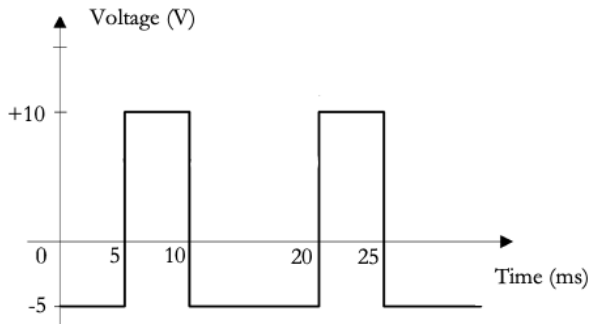
In the circuit below, the Zener diodes have a breakdown voltage of 5 V and $V_{DO} = 0.7$ V. The input signal is a sinusoid of amplitude 8 V.

d) Draw the output waveform with labels.
e) Draw the circuit transfer function with labels.



Consider the standard half-wave rectifier with a load resistor of 10 $k\Omega$ connected to it. The input to the rectifier is the periodic signal shown below. Use the diode ideal model.

d) Draw the rectified output waveform. Label all important points.
e) Find the average rectified output and the average load current.
f) If a capacitor of 470 μF is connected to the rectifier, how much is the ripple voltage. Draw the output waveform for this case (on top of the figure in a).



In the circuit below, the transistor has $|V_t| = 2$ V, k=0.2 mA/V^2, and λ=0. Find the value of the DC source V_{GG} such that the voltage at the drain terminal is 4 V. Show all your work.

